

Week 4 Tasks

Final Technical Presentation

5G Architecture

- **gNB (Next Generation NodeB):**

Provides radio connectivity between the User Equipment (UE) and the 5G Core.

Manages radio resources, scheduling, and handovers.

Ensures high-speed, low-latency wireless communication.

- **AMF (Access and Mobility Management Function):**

Handles user registration and authentication

Manages mobility and connection setup

Maintains UE connectivity within the network.

5G Architecture

- **SMF (Session Management Function):**

Establishes, modifies, and releases PDU sessions.

Assigns IP addresses to user devices

Selects the appropriate UPF for data routing

- **UPF (User Plane Function):**

Routes user data packets between the 5G Core and the Internet

Performs packet forwarding and traffic routing

Delivers high throughput with low latency.

- **5G Architecture Flow:**

UE-->gNB-->AMF-->SMF-->UPF-->Internet

Radio Technologies

- **OFDM (Orthogonal Frequency Division Multiplexing):**

Divides the available bandwidth into multiple orthogonal subcarriers.

Reduces interference and improves spectral efficiency.

Supports high-speed and reliable data transmission in 5G.

- **Beamforming:**

Directs radio signals toward a specific user instead of broadcasting in all directions.

Improves signal strength coverage, and network capacity

Reduces interference with neighboring users

Radio Technologies

- **Massive MIMO (Multiple Input Multiple Output):**

Uses a large number of antennas (eg., 32T32R, 64T64R)

Serves multiple users simultaneously using spatial multiplexing

Increases throughput, capacity, and spectral efficiency while improving coverage.

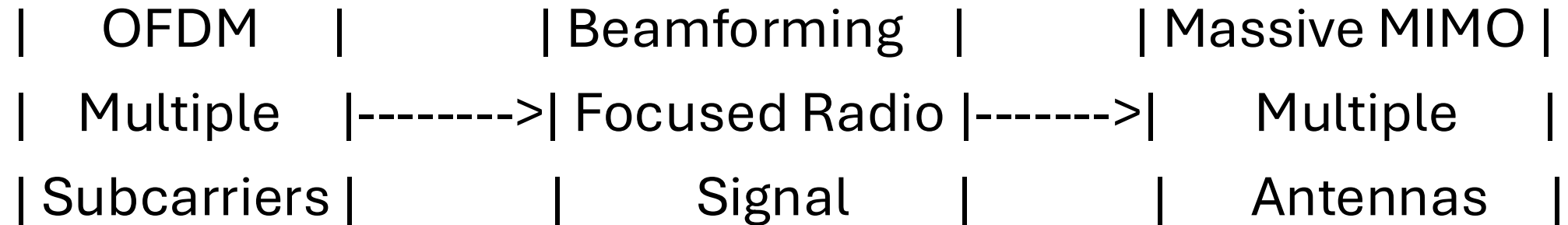
- **Key Benefits:**

OFDM: Efficient spectrum utilization and high data rates

Beamforming: Better signal quality and lower interference

Massive MIMO: Higher network capacity and improved user experience

Radio Technologies



Network Slicing

- Network Slicing is a 5G technology that creates multiple virtual networks (slices) on a single physical network.
- Each slice is optimized for specific applications and service requirements.
- Enables efficient resource allocation improved performance, and enhanced security.
- **Types of Network Slices:**

eMBB (Enhanced Mobile Broadband):

High-speed data services

Supports HD/4K video streaming, AR/VR, and cloud gaming

Focuses on high bandwidth and user experience

Network Slicing

URLLC (Ultra-Reliable Low-Latency Communication):

Provides ultra-low latency and high reliability

Used for autonomous vehicles, remote surgery, and industrial automation

Ensures real-time communication

mMTC (Massive Machine-Type Communication):

Supports millions of connected IoT devices

Designed for smart cities, smart agriculture, and environmental monitoring

Optimized for low power consumption and scalability

Network Slicing

- **Key Benefits:**

Customized network performance for different applications

Better resource utilization

Improved security and reliability

Supports diverse 5G use cases simultaneously

O-RAN

- O-RAN is an open and Intelligent Radio Access Network architecture. It enables interoperability between equipment from different vendors using open interfaces. Reduces deployment costs, increases flexibility, and encourages innovation.
- **Key Components:**
 - O-RU:** Handles radio transmission and reception
 - O-DU:** Performs real-time processing of radio protocols
 - O-CU:** Handles higher-layer protocol functions
 - RIC:** Used AI/ML to optimize network performance

KPI Analysis

- Key Performance Indicators (KPIs) measure the health and performance of a telecom network.
- They help operators monitor network quality and identify issues affecting user experience
- **Important KPIs:**

RSRP (Reference Signal Received Power): Measures the strength of the received signal

SINR (Signal-to-Interference-plus-Noise Ratio): Indicates signal quality by comparing signal strength to interference and noise

Throughput: Measures the amount of data transmitted per second

Latency: Measures the delay in data transmission (ms)

Packet Loss: Represents the percentage of data packets lost during transmission

KPI Performance Thresholds

KPI	GOOD	WARNING	CRITICAL
RSRP	>-90 dBm	-90 to -105 dBm	<-105 dBm
SINR	>20 dB	10-20 dB	<10 dB
Throughput	>100 Mbps	50-100 Mbps	<50 Mbps
Latency	<20 ms	20-50 ms	>50 ms
Packet Loss	<1%	1-3%	>3%

Packet Analysis

- Packet Analysis is the process of capturing and examining network packets.
- It helps monitor network traffic, troubleshoot issues, and verify protocol communication.
- Commonly performed using **Wireshark**.

- **Protocols Analyzed:**

ARP (Address Resolution Protocol): Maps IP addresses to MAC addresses

ICMP (Internet Control Message Protocol): Used for diagnostics and connectivity testing.

TCP(Transmission Control Protocol): Uses a three-way handshake (SYN-->SYN-ACK-->ACK)

Packet Analysis

UDP (User Datagram Protocol): Provides fast, connectionless communication

DNS (Domain Name System): Translates domain names into IP addresses

HTTPS/TLS: Encrypts data exchanged between clients and servers

- **Packet Analysis Workflow:**

Capture Packets-->Apply Protocol Filters-->Inspect Packet Details
-->Analyze Communication-->Identify Network Issues

Case Study Findings

- **Scenario:** A telecom cell was analyzed using the following KPIs:
- RSRP: -108dBm
- SINR: 8dB
- Throughput: 35 Mbps
- Latency: 90 ms
- Packet Loss: 5%
- **Key Findings:**

Weak signal strength resulted in poor network coverage.

Low SINR indicated high interference and reduced signal quality

High latency caused delays in real-time applications.

Case Study Findings

Increased packet loss led to unstable connections and retransmissions

Low throughput reduced data transfer speeds and overall user experience

- **Root Causes:**

Poor radio coverage

Network congestion

Signal interference

Backhaul capacity limitations

Suboptimal antenna configuration

Case Study Findings

- **Recommended Solutions**

Optimize antenna tilt and beamforming

Increase network capacity using Massive MIMO

Reduce interference through frequency optimization

Upgrade backhaul infrastructure

Continuously monitor KPIs for proactive maintenance